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Simple Line Editor in C

Team Members:-

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Problem understanding:

we have to design and build a command line-line editor.
c. The editor works with a text document line by line.
The user enters the commands through terminal to create view and modify the document.

Data Structure:

we choose an Array of strings

Declaration:

```
char lines[MAX_LINES][MAX_LENGTH];
```

```
MAX_LINES = 100
```

```
MAX_LENGTH = 200
```

Justification:

We choose array of strings because the editor handles a small text document. It's simple to implement and understand. Insertion and deletion can be performed by shifting the existing lines.

Commands:

1. insert <line_number> <text>
2. delete <line_number>
3. display
4. save <filename>

5. Load <filename>
6. help
7. Exit

Flowchart of the program:

Start

Initialize
LinesCount = 0

Display prompt
Read Command

Identify
Command

INSERT

DELETE

DISPLAY

Functions

Functions

Functions

More Commands

YES

NO

Read Command END

To implement Save/Load, add:

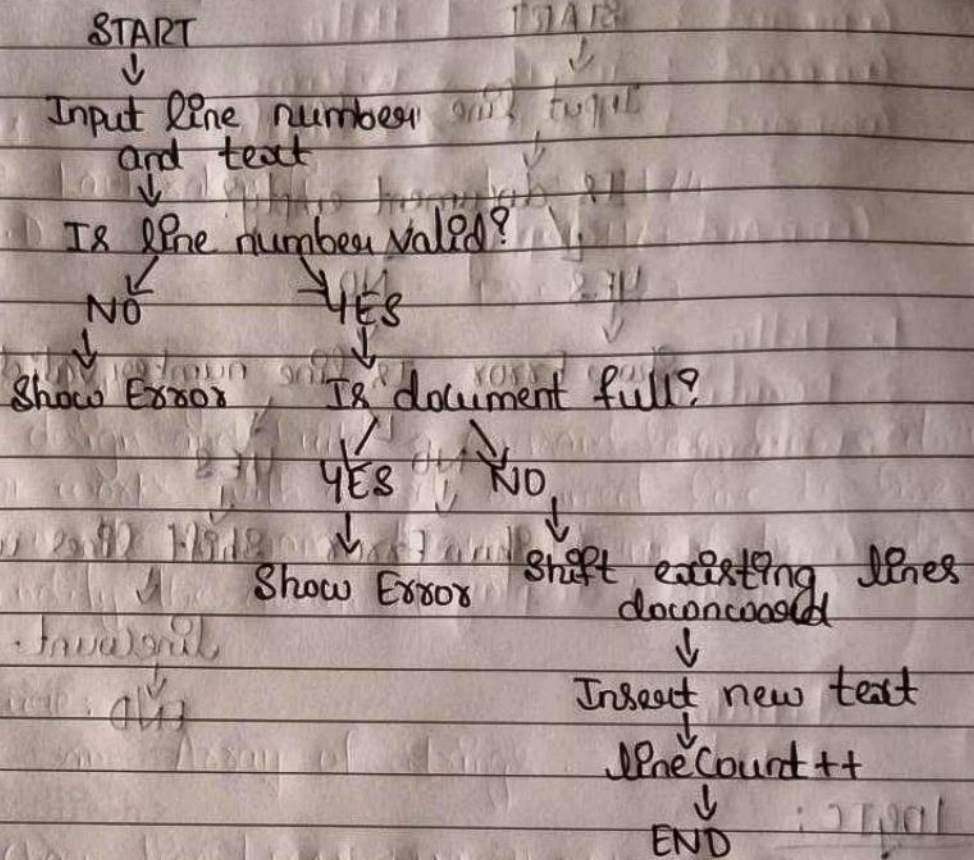
Save

Functions

Back to Command

COMS

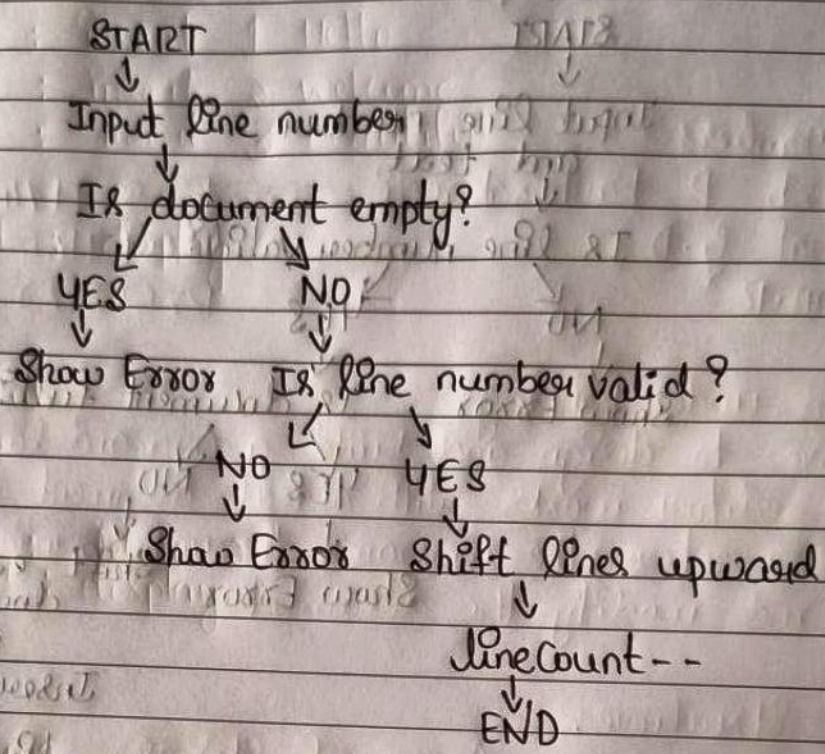
Insert line - Core logic :



Logic:

1. Read line number and text.
2. Check whether line number is valid.
3. Check whether the array has space.
4. Shift existing lines one position downwards.
5. Insert the new text.
6. Increase lineCount.

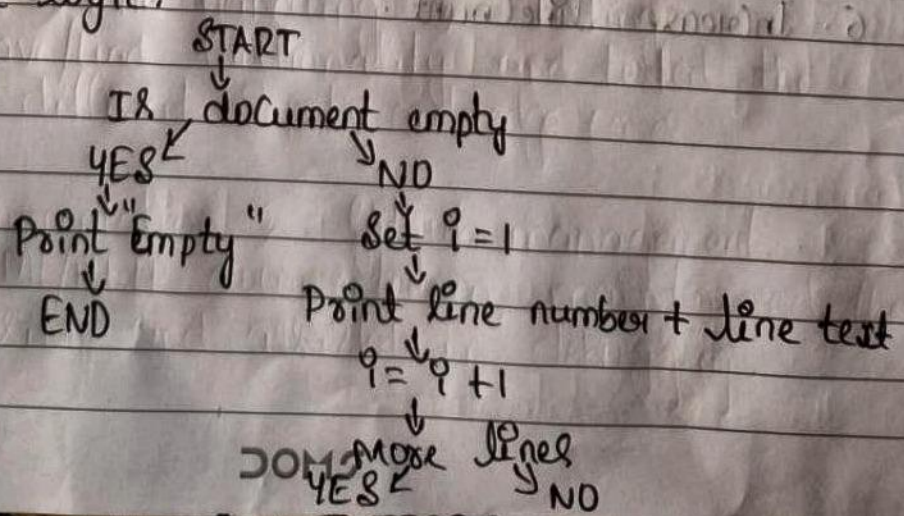
DELETE LINE - Core logic: : Signal, mail - mail, forward



LOGIC:

1. Read line number
2. Check whether document is empty
3. Check whether line number is valid
4. Shift all lines below the selected line upwards.
5. Decrease linecount.

DISPLAY - Core logic:



NO
↓
END

Logic:

1. Check whether document is empty.
2. If empty, display a suitable message.
3. Otherwise start from line 1.
4. Print line number and text.
5. Move to the next line.
6. Continue until all lines are displayed.

PROGRAM:

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
```

```
#define MAX_LINES 100
#define MAX_LENGTH 256
```

char document[MAX_LINES][MAX_LENGTH];

Function Design:

- main() → Controls the program.
- insertLine() → Inserts a new line.
- deleteLine() → Deletes a line.
- displayDocument() → Displays all lines.
- saveFile() → Saves document to file.
- loadFile() → Loads document from file.
- Show help() → Displays available commands.

EXAMPLE:

Command : Insert 1 Hello

Command : Insert 2 Welcome

Command : Insert 3 Beautiful

Command : display

1. Hello

2. Welcome

3. Beautiful

Command : delete 2

Command : display

1. Hello

2. Beautiful

Team Members and Roles:

Vighnesh P Kanhakarandi - Team leader and
Core logic developer.

Designed the overall program structure and implemented
insert, delete and display functions.

Ujjwal G.D - File handling & Testing
Implemented save/load functionality and tested the
program for different inputs and edge cases.

Rohan.R - Documentation & Github management
prepared README/HELP documentation, maintained the
github repository and assisted in debugging.

DOMS